

PecoDrop Reverse Vending Machine

OPERATOR & CUSTOMER USER MANUAL • 3-CHAMBER SMART KIOSK

DOC REF: RVM-OP-2.4.0

HARDWARE: Mega 2560

Pro

DATE: 2026-09-16

01 System Overview & Product Architecture

The **PecoDrop 3-Chamber Reverse Vending Machine (RVM)** is an automated electromechanical kiosk engineered for high-throughput collection, automated sorting, volumetric sizing, material purity verification, and instant digital reward disbursement for consumer recyclables.



Figure 1.1: PecoDrop 3-Chamber Smart Kiosk with Central Interactive Touchscreen & Illuminated Intake Ports

Chamber 1 (Cyan Glow)

PET Plastic Beverage Bottles. 3-tier vertical ultrasonic array for Small ($\leq 250\text{ml}$), Medium (500ml), and Large (1.5L+).

Chamber 2 (Amber Glow)

Beverage Aluminum & Steel Cans. Inductive proximity sensor executing 12-sample burst over 180ms (≥ 9 LOWs = 100% metal).

Chamber 3 (Emerald Glow)

Clean Paper & Cardboard. Dual ultrasonic arrival check and HX711 24-bit ADC strain-gauge scale with dynamic tare calibration.

02 Customer Recycling Deposit Workflow

Follow the 4–step interactive procedure to deposit items and earn instant digital Green Points:

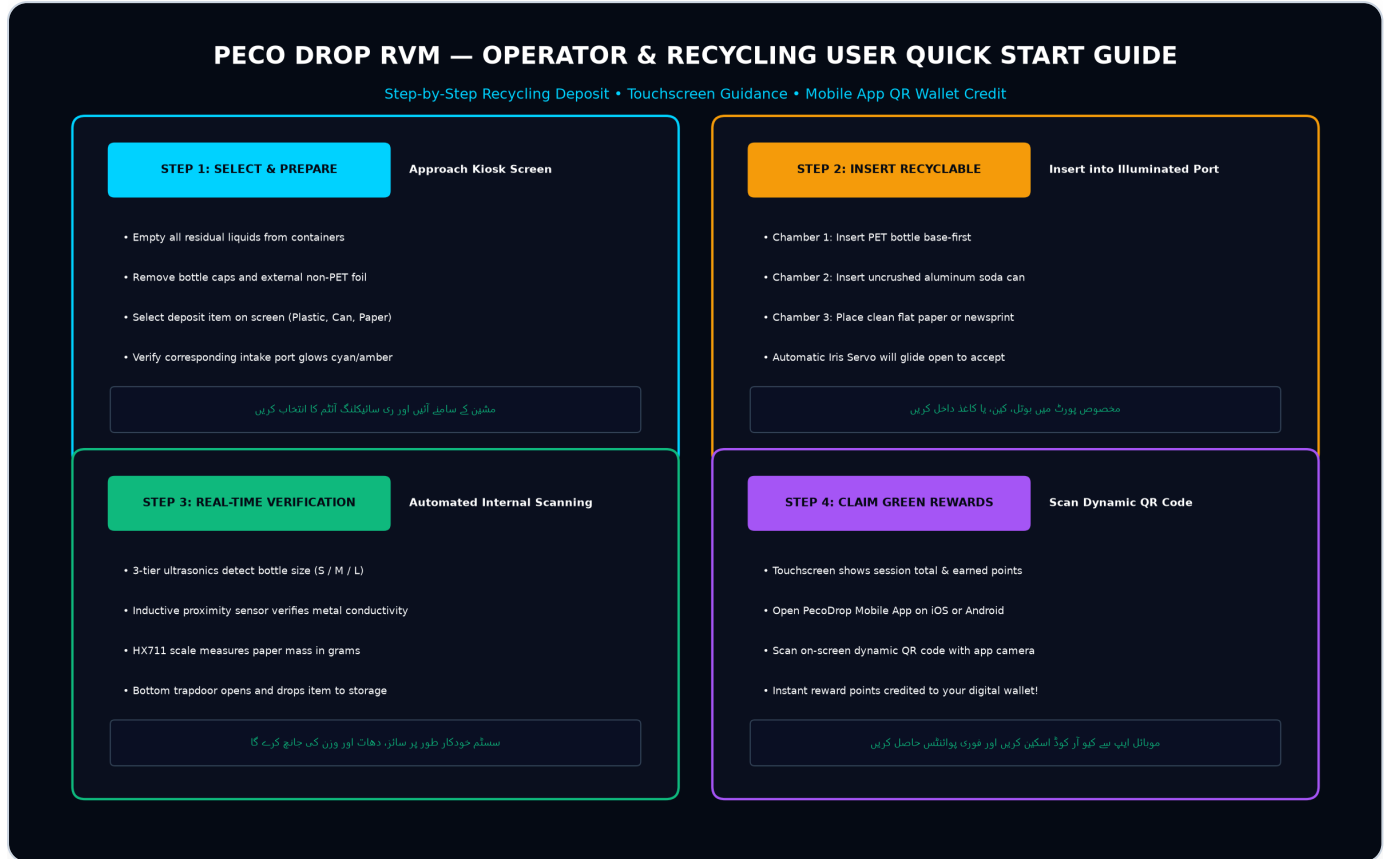


Figure 2.1: 4–Step Customer Recycling Workflow & Mobile App Dynamic QR Code Reward Claim Guide

STEP 1 Approach Kiosk & Select Waste Category

Stand before the kiosk. On the touchscreen, select **Plastic Bottles**, **Aluminum Cans**, or **Paper**. The designated intake port softly illuminates.

مشین کے سامنے آئیں اور اسکرین پر اپنی ری سائیکلنگ کیٹیگری کا انتخاب کریں۔

STEP 2 Empty Container & Insert Base–First

Ensure containers are empty of liquids. Insert into the active port. The automatic upper iris aperture gate glides open (10° to 120°) to accept the item safely.

مخصوص پورٹ میں بوتل، کین، یا کاغذ داخل کریں۔

STEP 3 Automated Internal Sizing & Material Verification

Iris closes. The internal array scans: height ultrasonic beams for bottles, inductive sensor for aluminum purity, or load cell for paper mass. The bottom drop gate discharges into the internal hopper.

مشین خودکار طور پر سائز، دھات اور وزن کی تصدیق کر کے ذخیرہ کرے گی۔

STEP 4 Scan Dynamic QR Code & Claim Green Points

Session summary appears on screen. Launch the **PecoDrop Mobile App**, tap *Scan QR*, and scan the on-screen code to credit points immediately to your digital wallet!

موبائل ایپ سے اسکرین پر موجود کیو آر کوڈ اسکین کریں اور فوری انعامات حاصل کریں۔

03 Accepted vs. Rejected Items Matrix

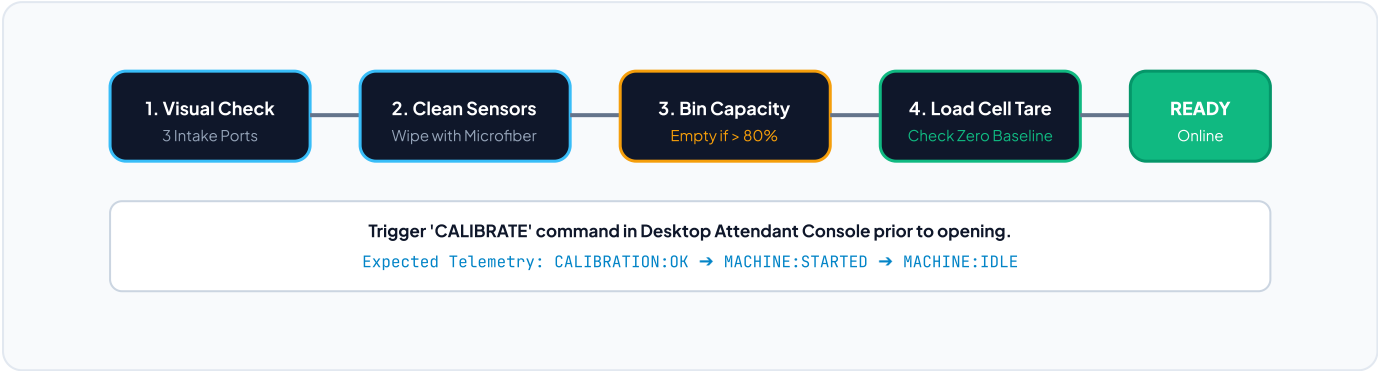
RECYCLING STREAM	INTAKE PORT	ACCEPTED ITEMS	STRICTLY REJECTED ITEMS
PET Plastic Bottles	Chamber 1 (Cyan Glow)	• Clear & tinted PET bottles (250ml – 2.5L) • Water, soft drink & juice bottles • Clean empty containers	• Containers with liquid contents • Crushed or flattened bottles • Glass bottles, motor oil cans • PVC pipes and household plastics
Beverage Cans	Chamber 2 (Amber Glow)	• Standard 330ml / 500ml aluminum cans • Slim 250ml energy drink cans • Clean tinplate food beverage cans	• Severely crushed/shredded metal foil • Pressurized aerosol spray cans • Plastic coated metal composites • Batteries or electronic scrap
Paper Wastage	Chamber 3 (Emerald Glow)	• White office paper, letters, notebooks • Newspapers, magazines & brochures • Clean flattened cardboard packaging	• Wet, grease-soaked, or food-soiled paper • Thermal receipts & carbon paper • Styrofoam & plastic bubble wrap • Heavy metal binders & large clips

04 Bilingual Operational Prompts (English & Urdu)

SYSTEM STATE	ENGLISH DISPLAY BANNER	URDU AUDIO/VISUAL GUIDANCE (اردو)
Kiosk Ready	<i>System Ready. Please insert plastic bottle, can, or paper.</i>	مشین تیار ہے • برائے مہربانی پلاسٹک کی بوتل، کین، یا کاغذ داخل کریں
Item Accepted	<i>Object Accepted. Scanning size & material purity...</i>	آئٹم قبول کر لیا گیا • سائز اور میٹریل کی جانچ جاری ہے
Plastic Sized	<i>Plastic Bottle verified: Medium Tier (+15 Points).</i>	پلاسٹک کی بوتل تصدیق شدہ: درمیانی سائز (15+ پوائنٹس)
Metal Verified	<i>Aluminum Can verified: 100% Metal (+20 Points).</i>	ایلومینیم کین کی تصدیق مکمل (20+ پوائنٹس)
Paper Weighed	<i>Paper mass logged: 0.185 kg (+25 Points).</i>	کاغذ کا وزن کامیابی سے ریکارڈ کیا گیا (25+ پوائنٹس)
Session Finished	<i>Session Complete! Scan dynamic QR code on screen.</i>	سیشن مکمل! اسکرین پر موجود کیو آر کوڈ اسکین کریں
Error / Jam	<i>Chamber intake blocked. Attendant alerted.</i>	چیمبر میں رکاوٹ • عملے کو مطلع کر دیا گیا ہے

05 Attendant Daily Operational Checklist

To ensure high availability and prevent jams, attendants must execute this workflow daily:



Morning Opening Routine

- 1. **Power Supply:** Ensure primary 12V 15A industrial PSU is energized and green LEDs are illuminated.
- 2. **Servo Gate Verification:** Confirm all 3 iris gates and 3 drop gates are in the secure closed position (10°).
- 3. **Sensor Lens Cleanliness:** Wipe ultrasonic transducer cones and inductive sensor face with antistatic microfiber cloth.
- 4. **App Connectivity:** Verify host PC running PecoDropDesktopApp has opened COM port at 115200 baud.
- 5. **Auto-Calibration:** Send CALIBRATE. Confirm CALIBRATION:OK and MACHINE : IDLE appear on logs.

EMERGENCY PROCEDURES

- **Object Jam:**
Press the red EMERGENCY STOP button on the kiosk side or send STOP via console. All servos lock to 10° immediately.
- **Liquid Ingress:**
Immediately trip the main safety breaker switch located on the bottom-right interior cabinet.